

Differential Climate Impacts on Fishing Effort in Chilean Small Pelagic Fisheries

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Abstract

Climate-driven shifts in stock productivity reach fishing fleets unevenly when species portfolios differ across fleet segments and the sector-level allocation of catch rights cannot adjust to changes in relative productivity. We quantify this distributional channel for the multi-species small pelagic fishery of Central-South Chile (anchoveta, sardine, jack mackerel), managed under sector-specific quota regimes with a fixed industrial–artisanal allocation. Our framework couples a Bayesian state-space model of stock dynamics, fit to 2000–2024 official assessment data, with a vessel-level trip equation; climate enters through an indirect biomass channel and a direct weather channel. Under a six-model CMIP6 ensemble spanning two emissions scenarios and two time windows, projected fleet-level effort falls by 13–15 percent for the artisanal fleet but only 0.8–1.0 percent for the industrial fleet — an asymmetry of roughly fifteen to one driven by differential portfolio exposure and weather sensitivity, with portfolio exposure decomposed between fleets and within the artisanal fleet by vessel type. The findings imply that climate-adaptation policy in Chile should revisit the sector-level allocation of catch rights, allowing it to respond to relative productivity shifts rather than remain fixed independent of the climate state.

Keywords: Bayesian state-space model; Bioeconomic projection; Chilean small pelagic fishery; Climate change; Fishing effort; Fleet heterogeneity; Quota allocation

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1 Introduction

Climate change is reshaping the productivity of marine fisheries, but the incidence of this reshaping across fleet segments is rarely uniform. Quota-allocation regimes that codify the relative shares of industrial and artisanal (small-scale) fleets were typically designed under stationary-climate assumptions, so a fleet-asymmetric productivity shock acts as an unscripted reallocation of harvest rights. Whether such regimes should anticipate this reallocation depends on the projected magnitude and asymmetry of fleet-level effort responses under credible climate scenarios—a quantity the empirical literature has yet to deliver for most major multi-species fisheries.

We provide that quantity for the small pelagic fishery (SPF) of Central-South Chile (Regions V–X and XIV), part of the country’s largest fishery—approximately 86% of national fish landings by industrial and artisanal sectors in 2019 (?)—and the segment in which all three target species, anchoveta (*Engraulis ringens*), sardina común (*Strangomera bentincki*), and jack mackerel (*Trachurus murphyi*), coexist in non-negligible shares of both fleet portfolios. These three species share habitat and are coupled through trophic and market linkages (??; ?). The fishery is managed through a sector-specific quota architecture: a fixed industrial-versus-artisanal allocation splits the Total Allowable Catch between the two segments, while within-sector distribution operates through Transferable Fishing Licenses (*Licencias Transables de Pesca*, LTP) for the industrial fleet and the *Régimen Artesanal de Extracción* (RAE) for the artisanal fleet, with limited cross-sector transferability. Because the two segments target different species portfolios, a stock-asymmetric climate shock translates directly into a fleet-asymmetric effort response. Cross-country evidence indicates that Chile is projected to lose part of its maximum catch potential under climate change: Cheung et al. (?) classify Chile among a group of countries with moderate projected declines (6–13% by 2055 under SRES A1B), including the contiguous United States, China and Brazil; the available climate-fisheries economics literature is patchy globally and concentrated in a few well-studied regions (?). To our knowledge, no existing study for Chile combines multi-species stock dynamics with vessel-level effort responses under projected climate scenarios.¹

Our bioeconomic framework combines two empirical components. First, a Bayesian state-space model of multi-species stock dynamics — fit to 2000–2024 official IFOP and SPRFMO assessments — estimates, for each species, two semi-elasticities of productivity with respect to sea-surface temperature and log-chlorophyll-a anomalies. We call these parameters *climate shifters*. Because they enter the biological law of motion as coefficients on exogenous climate forcings, they remain interpretable when applied to climate states outside the 2000–2024 sample. Second, a vessel-level negative binomial trip equation — estimated separately for the industrial and artisanal fleets, with year fixed effects that absorb aggregate non-climate shocks (notably the 2019–2020 social and pandemic disruptions) — maps prices, allocated harvest, weather, and regulatory closures into annual fishing effort. Climate reaches fleet-level effort through two channels: an *indirect*

¹The only empirical study of Chilean fisher behavior that we are aware of is Peña-Torres et al. (?), which analyses how the El Niño–Southern Oscillation (ENSO) affects location choices in the jack mackerel fishery.

biomass channel via the climate shifters, and a *direct weather channel* via vessel-specific exposure to severe winds. We evaluate the framework against a six-model CMIP6 ensemble under SSP2-4.5 and SSP5-8.5, at mid-century (2041–2060) and end-of-century (2081–2100). Throughout, we report two sources of uncertainty separately: dispersion across climate models (which future climate trajectory we project against) and dispersion in the Bayesian posterior of the structural parameters (uncertainty about the elasticities themselves)

We obtain three sets of results. First, the distributional incidence of climate-driven productivity shocks across fleets is sharply asymmetric: the artisanal fleet contracts by 13.2–15.0% across CMIP6 scenarios, while the industrial fleet contracts by only 0.8–1.0% — a ratio of roughly fifteen to one. Two economic mechanisms compound to produce this asymmetry. Each fleet is exposed to climate-driven productivity shifts through a distinct species portfolio, and the fixed industrial–artisanal allocation binds, preventing the cross-sector reassignment of harvest rights that would otherwise equalize marginal returns under the new relative productivities. The result instantiates, in a climate-shock setting, the portfolio-and-access framework of Kasperski & Holland (2019) and Cline et al. (2020): segments with narrower species portfolios bear sharper variance, and the welfare cost of that exposure rises when institutional access constraints prevent rebalancing (2019). Because adaptation responses at the individual fisher level are themselves heterogeneous and context-dependent (2020; 2021), the policy reading is direct: cross-sector transferability is the first-order rent-reallocation margin for attenuating disproportionate fleet-level losses, and any reform—the size, direction, and shadow price of admissible transfers—is quantifiable from the posterior we report. Second, the long-run change in intrinsic productivity is large and negative for anchoveta and sardine—the two coastal-upwelling stocks that dominate the artisanal portfolio—with floor effects for sardine under SSP5-8.5 end-of-century. Third, the climate shifters are sharply identified for these two stocks but not for jack mackerel, the species that anchors the industrial portfolio. We document this null with four converging tests (spatial-domain robustness across nested coastal aggregations, dual-source biomass with the Northern Chilean acoustic series, an OROP-PS coherence diagnostic, and basin-scale ENSO Niño 3.4 as an alternative shifter) and treat the jack mackerel productivity factor as fixed at its prior mean throughout the projections.

The remainder of the paper is organized as follows. Section 2 describes the economic and institutional setting of the Chilean SPF. Section 3 presents the data, the stock-dynamics model, the trip equation, and the CMIP6 projection design (Section 3.1). Section 4 reports the identified climate shifters, the comparative-statics projections of stock productivity, and the projections of fleet-level effort (Section 4.1). Section 5 discusses policy implications and limitations; Section 6 concludes.

2 The small pelagic fishery in Chile

The Chilean small pelagic fishery (SPF) is the largest fishery in the country, accounting for approximately 86% of national fish landings by the industrial and artisanal sectors in 2019 (?). The fishery is divided into two latitudinal zones with distinct species compositions. The northern zone (Regions I–IV) is dominated by anchoveta (*Engraulis ringens*) and jack mackerel (*Trachurus murphyi*), while sardine (*Strangomera bentincki*) is effectively absent. In contrast, the Central-South zone (Regions V–X and XIV) includes all three species in non-negligible shares within both artisanal and industrial landings. Because this region supports a genuinely multi-species fishery operating under a fixed sector-level allocation regime, it provides the relevant setting for evaluating the distributional consequences of climate-induced shifts in stock productivity, and is therefore the focus of this paper. The principal ports servicing this zone include San Antonio, Tomé, Talcahuano, San Vicente, Coronel, Lota, and Corral. The three species are harvested primarily by purse-seine fleets that differ in vessel size, operating range, and species composition.

The historical evolution of the fishery illustrates the portfolio mechanism developed in the remainder of the paper. The jack mackerel fishery was initially concentrated in northern Chile but shifted toward the Central-South zone during the mid-1980s (?), where fishing grounds have traditionally operated within 50 nautical miles of the coast. Between 1987 and 2004, approximately 85% of jack mackerel landings were directed to the reduction industry (?), reflecting the importance of fishmeal and fish-oil export markets in determining the value of the industrial portfolio. Sardine and anchoveta, by contrast, are jointly harvested using the same gear and fishing grounds. The two species share habitat and fishing technology, making species differentiation at the point of capture effectively impossible (?). As a result, substitution between them is constrained primarily by quota regulations rather than by biological or technological limitations.

The status of the three stocks has evolved substantially over time. Central-South anchoveta was classified as collapsed through 2018, reclassified as overexploited in 2019, and has been harvested within maximum sustainable yield (MSY) limits since 2020. Sardine has generally remained within MSY reference levels, except in 2021 and 2023 when it was classified as overexploited. Jack mackerel remained overexploited until 2018 and has since been harvested within MSY limits.

The Chilean SPF is managed through an annual Total Allowable Catch (TAC; *Cuota Global*) that is divided between the industrial and artisanal sectors, with small fractions reserved for research and contingency purposes. TACs are further subdivided across regions and seasons, and unused quotas may be reassigned within the fishing year. Anchoveta and sardine are managed under a mixed-species quota system: although each species has an independent quota, substitution between them is permitted within the regulatory framework.

Since 2013, the industrial sector has operated under an individual transferable quota (ITQ) system known as Transferable Fishing Licenses (*Licencias Transables de Pesca*, LTP). Class A licenses were allocated according to historical catch shares, whereas Class B licenses—representing

up to 15% of the industrial allocation—have been distributed through sealed-bid first-price auctions since 2015. Peña-Torres et al. (?) document that these auctions have exhibited low participation, limited capacity to reflect economies of scale, and indications of coordinated bidding behavior. The artisanal sector operates under a regulated open-access regime, with regional TACs distributed through the *Régimen Artesanal de Extracción* (RAE) according to area, vessel size, landing site, or fisher organization.

Biological closures (*vedas*) constitute a central management instrument for sardine and anchoveta. In the Central-South zone, reproductive closures extend from July to October in Regions IX–XIV and from August to October in Regions V–VIII, while recruitment closures occur from January through February across all regions. In total, these regulations generate 151 closed fishing days per year in Regions V–VIII and 182 days in Regions IX–XIV. Jack mackerel is not subject to seasonal biological closures.

Beyond the within-sector quota architecture described above, the partition of each Total Allowable Catch between the artisanal and industrial sectors is fixed by statute at the legislative level. Article~47 of the General Law of Fisheries and Aquaculture (Ley General de Pesca y Acuicultura, LGPA) governed this de jure partition during the estimation window of the present paper (2018–2024); a 2025 reform, Ley~21.752, fixes a new partition in force until 2040. Table~?? summarises the two regimes for the three Central-South small pelagic stocks of interest. Under the pre-2025 regime, the artisanal sector held 78% of the Central-South anchoveta and sardina com'un TACs and 10% of the Central-South jurel TAC across the V–Los~Lagos macrozone; Ley~21.752 raises the artisanal share to 90% in both small pelagic stocks and partitions jurel into a 30% artisanal share in V–Los~R'ios and a 15% share in Los~Lagos, for a quota-weighted aggregate of approximately 28% across the V–Los~Lagos macrozone. The de jure partition admits a single inter-sectoral re-allocation channel, governed by Article~55T of the LGPA: industrial titulares and artisanal armadores may cede assigned tonnage across the sectoral boundary subject to ex-ante SUBPESCA authorisation, with the transferred volume drawn from the cedent's residual quota and added to the cessionary's effective quota for the calendar year. The SERNAPESCA quota control reports for 2013–2024 document that this channel is heavily used in the Central-South small pelagic fishery (Appendix~??, Section~?): the industrial sector cedes a stable 85–99% of its assigned anchoveta and sardina com'un TACs to artisanal cessionaries every year, in volumes ranging from 25 to 77~kt per species-year, while jurel cessions are an order of magnitude smaller and shift direction across years. Two consequences for the present paper follow directly. First, the empirical specifications in Sections~?? and~?? are identified on realised landings by sector, which are post-cession; the asymmetric climate sensitivity we report therefore reflects the operational allocation produced by the joint Article~47–Article~55~T regime, not the de jure Article~47 fractioning alone. Second, Article~6 of Ley~21.752 imposes a specific patent of 1.3 monthly tax units per transferred ton on industrial-to-artisanal cessions, with artisanal armadores exempt from the patent in either direction; this is a transactional cost on the pre-existing Article~55T channel rather than the introduction of a new wedge. Within sectors, transferability also differs sharply: industrial coefficients trade under the